



ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your Post-Viral *Recovery*, Ultimate Guide

*A Functional Medicine Guide to Immune Activation, EBV Reactivation,
and Recovering From Long COVID and Post-Viral Syndromes*

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

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BEFORE YOU START

A note before you start. This guide is educational information. It explains, in plain language, the immune and cellular mechanisms behind why some people do not fully bounce back after a viral illness such as COVID-19 or Epstein-Barr virus (EBV, the virus that causes mono), and the general categories of functional medicine support used to address those mechanisms. It is not individualized medical advice, and it does not replace a relationship with your physician or a clinical evaluation with my team. Never stop or change a prescribed medication without talking to the prescriber first. This guide does not diagnose Long COVID, chronic EBV, or any other condition. Talk to your doctor or a qualified practitioner before starting any new supplement, especially if you take blood thinners, heart medications, or other prescriptions, and see the red flag section before anything else if your symptoms are severe, new, or rapidly worsening.

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When the Illness Is Over, but You Are Not Better

You had the virus. Maybe it was COVID-19. Maybe it was mono, the flu, or something you never got a name for, just a rough week that never quite ended. Your test eventually came back negative, or you were never tested at all. Everyone around you moved on. You did not.

Months later you are still tired in a way that sleep does not fix. Your brain feels like it is working through fog. Your heart races when you stand up. You get sick again more easily than you used to, or you develop new sensitivities to foods, smells, or medications that never bothered you before. You go to your doctor, the standard bloodwork comes back essentially normal, and you are told that everything looks fine. You do not feel fine.

This experience is genuine and common, and it now has a substantial and growing scientific literature behind it. It is not a character flaw or “just anxiety,” and you are not imagining it. What it usually is, is a nervous system, an immune system, and a cellular energy system that took a genuine hit from an infection and have not yet finished recalibrating, sometimes because the virus or its remnants are still present in some form, sometimes because the immune activation the virus triggered has outlasted the virus itself.

Here is a fact that reframes how seriously this deserves to be taken. In 2022, a landmark study followed more than ten million young adults in the United States military and found that the risk of developing multiple sclerosis, one of the most feared neurological diseases there is, increased 32-fold after infection with EBV, and did not increase after infection with other common viruses, including the similarly transmitted cytomegalovirus.¹ The researchers concluded that these findings point to EBV as a leading cause of MS. If a virus that the great majority of adults carry without ever knowing it can be a leading driver of a disease as serious as multiple sclerosis, decades after the original infection, then the idea that a virus can leave a lasting mark on your immune system, your energy, and your neurological function is an established principle of modern virology rather than a fringe theory, and it is the foundation this entire guide is built on.

This guide exists for two groups of people, and their experiences overlap heavily: those recovering from COVID-19 whose symptoms never fully resolved, sometimes called Long COVID, and those whose fatigue, brain fog, and other symptoms trace back to EBV, the virus behind mono, that never seems to have fully quieted down. It walks through nine keys: what it actually means when an infection “does not fully leave,” how immune activation drives the symptoms you are feeling, what EBV blood testing does and does not tell you (this is the single most misunderstood topic in this entire field, and getting it right protects you from both false reassurance and false alarm), the current science and its limits on whether virus fragments persist in the body, how the gut and immune system stay entangled with each other during recovery, the mitochondrial and energy side of the picture, the autonomic nervous system symptoms like a racing heart on standing, the mast cell and histamine overlap that makes everything feel reactive, and what a genuinely useful functional medicine workup looks like. It closes with a staged roadmap, a self-tracking worksheet, and the honest limits of what is proven versus what is still hypothesis.

A note on scope. If a tick bite or time in a tick-endemic area is part of your history, *Borrelia* and its co-infections (*Babesia*, *Bartonella*, *Ehrlichia*) belong in the conversation, and my *Unlock Your Lyme* guide covers that territory in complete detail. This guide does not re-cover it. If recurring yeast or fungal overgrowth is your primary picture, the *Unlock Your Candida* guide is the better starting point. And if your fatigue, brain fog, and unrefreshing sleep are the dominant complaint with no clear post-viral trigger, or if mitochondrial and HPA-axis (stress hormone) mechanics are what you most want to understand, my *Unlock Your Energy* guide goes deep on that terrain. This guide covers the same fatigue and brain fog territory specifically through the lens of what a virus did to your immune system, so expect some overlap in symptoms and a deliberately lighter touch on the mitochondrial and stress-hormone chapters, which the *Energy* guide covers in full.

What “The Infection Never Fully Left” Actually Means

Two framings, both wrong. Patients are frequently told one of two things: either the virus is gone and nothing further should be wrong, or, less often, that they have “chronic” this or that infection with no clear explanation of what that word is supposed to mean biologically. Neither framing is quite right, and the vagueness is part of why this is so disorienting to live through.

What “never fully left” actually means. “The infection never fully left” can mean genuinely different things depending on the virus and the case, and part of recovering well is understanding which one applies to you. It can mean the immune system’s response to the original infection has not fully stood down, so you are dealing with the consequences of ongoing immune activation rather than an ongoing infection itself. It can mean a virus that is designed by its own biology to persist, like EBV, has reactivated from its normally dormant state more than usual. It can mean fragments of viral genetic material or protein remain detectable in tissue for longer than expected, without necessarily representing a live, replicating infection. And for a meaningful number of people, it can mean a new, separate illness pattern was triggered by the infection and is now sustaining itself through its own mechanisms (deconditioning, disrupted sleep architecture, autonomic dysregulation) even after the original immune trigger has faded. Most patients are living with some combination of these, not just one.

This is not new territory for medicine, even though COVID-19 brought it into the mainstream conversation. Long before COVID-19 existed, researchers already knew that a meaningful minority of people who catch an ordinary infection go on to develop a persistent fatigue syndrome. A 2006 prospective study out of Dubbo, Australia followed 253 patients from the moment of acute infection with one of three completely unrelated pathogens, EBV (glandular fever), Q fever (a bacterium), or Ross River virus (a mosquito-borne virus), and found that roughly 1 in 10 of them still met diagnostic criteria for chronic fatigue syndrome six months later, regardless of which of the three pathogens they had caught.² The syndrome that followed looked essentially the same no matter which organism triggered it, which is itself an important clue: the illness that follows may have less to do with the specific germ and more to do with how a given person’s immune system responds to a significant infectious insult.

COVID-19 has simply put a name and a number on what post-infectious illness researchers already suspected. The World Health Organization, through an international Delphi consensus process involving 265 patients, clinicians, and researchers, formally defined post-COVID-19 condition (commonly called Long COVID) as occurring in someone with a probable or confirmed history of SARS-CoV-2 infection, with symptoms that usually begin around three months from the onset of COVID-19, last for at least two months, and cannot be better explained by another diagnosis.³ A 2025 multidisciplinary compendium from over 40 Long COVID clinical centers across the United States estimates the current global prevalence at approximately 6 percent of people who have had COVID-19, with the most common features being fatigue and a diminished energy window, post-exertional malaise (a worsening of symptoms after physical or mental

exertion, covered in full in Key 6), cognitive impairment often called brain fog, autonomic symptoms, pain, and changes in smell or taste.⁴ Critically, that same compendium is explicit that there is currently no single laboratory test that definitively confirms or rules out Long COVID. It is a clinical diagnosis built from history and pattern, not a single blood value, which is exactly why the testing chapter later in this guide focuses on characterizing your terrain rather than chasing one number.

What this means for you. The label matters less than the mechanism. Whether your case is being driven primarily by lingering immune activation, viral reactivation, tissue-level viral remnants, or a self-sustaining secondary pattern determines which lever is most worth pulling first, which is exactly what the rest of this guide is built to help you understand.

Immune Activation and the Inflammatory Signature

A word marketing has emptied out. “Inflammation” gets used so loosely in wellness marketing that it has almost stopped meaning anything. In post-viral illness, it means something specific and measurable: an immune system that was appropriately activated to fight off a real threat, and has not fully returned to its resting baseline once the threat was handled.

How the curve should look, and how it looks instead. A healthy immune response to a virus looks like a bell curve. It ramps up quickly, does its job, and then stands down over days to a couple of weeks as the threat clears. In post-viral syndromes, a subset of patients show a flattened, extended version of that curve instead. A comprehensive 2023 review from the NIH’s RECOVER research initiative, examining the pathogenic mechanisms behind Long COVID, describes ongoing activity of primed immune cells, clotting and coagulation dysregulation, dysfunctional brainstem and vagus nerve signaling, and autoimmunity arising from molecular mimicry (where the immune system’s antibodies against a viral protein accidentally cross-react with the body’s own tissue) as parallel, overlapping contributors to the syndrome, alongside the persistent viral reservoirs and latent-virus reactivation covered in Keys 3 and 4.⁵ Rather than a single broken switch, this is several immune subsystems that have each drifted slightly out of their normal operating range, and the drift compounds.

One specific and well-documented example illustrates how far this drift can go. In patients hospitalized with life-threatening COVID-19 pneumonia, researchers found that at least 1 in 10 had developed autoantibodies, meaning antibodies the immune system had mistakenly generated against its own tissue, that specifically neutralized type I interferon, the family of signaling proteins the body relies on to mount an effective antiviral response in the first place.⁶ These autoantibodies were essentially absent in people with mild or asymptomatic infection. This is a clear, confirmed example of a virus provoking an immune response that turns against a piece of the body’s own antiviral machinery, and it is part of why a functional medicine workup for post-viral illness looks at broader immune-regulation markers, not just a single inflammatory number. It is worth being precise about what this particular finding does and does not show: it was identified in patients with severe acute pneumonia, not specifically in people with Long COVID or chronic EBV, and it is presented here as an illustration of how real and specific post-viral immune dysregulation can get, not as a claim that this exact mechanism explains every case of lingering fatigue.

Why there is no immune-boosting protocol here. This is why post-viral care in a functional medicine setting rarely starts with a single supplement aimed at “boosting immunity.” An immune system that is already overactivated in the wrong direction does not need a blunt stimulant. It needs its regulatory signals restored, which is a terrain question (gut, sleep, nervous system, nutrient status) more than a single-ingredient question, and it is why the chapters that follow build out that terrain piece by piece rather than jumping straight to a product list.

EBV Serology, What It Does and Does Not Tell You

This is the single most important chapter in this guide to get right, and if you remember only one thing from everything you read here, it should be this one.

What a positive result doesn't mean. A huge number of people are told, based on a blood test showing “positive” EBV antibodies, that they have a chronic or reactivated EBV infection driving their fatigue. This is frequently an oversimplification bordering on a misreading of the test, and understanding why is the difference between a useful workup and a misleading one.

How common EBV actually is. EBV is one of the most common viruses in humans. It is not a rare virus that only some people carry. Using nationally representative US health survey data, researchers found that by ages 18 to 19, 89 percent of young people in the United States already carry EBV antibodies, a figure that had been climbing steadily higher with each older age group in the same dataset (50 percent by ages 6 to 8, 55 percent by 9 to 11, 59 percent by 12 to 14, 69 percent by 15 to 17).⁷ In other words, the great majority of adults are EBV seropositive, and that is true whether or not they have ever felt sick from it a day in their life. This is the single most important fact for interpreting your own labs. A positive EBV IgG antibody result, by itself, tells you only that you were infected with EBV at some point, typically years or decades ago, exactly like almost everyone else. It does not, on its own, tell you that EBV is currently active, reactivated, or responsible for how you feel today.

Real EBV serology has several components, and reading them correctly requires looking at the pattern, not one number in isolation. Viral capsid antigen (VCA) IgM rises early in a first, acute infection and then disappears, so a positive VCA IgM usually signals a genuinely new or recent infection. VCA IgG rises during that same acute infection, then stays positive for life in almost everyone, which is the “everyone has this” antibody described above. EBNA (Epstein-Barr nuclear antigen) IgG typically does not appear until weeks to months after a first infection and then also persists for life, and its presence is one of the ways labs distinguish a truly recent first infection (EBNA still negative) from a past one (EBNA positive). Early antigen-diffuse (EA-D) IgG is the marker most associated, though imperfectly, with reactivation, since it is more often present or elevated during periods of active viral replication and tends to fade between reactivation episodes.

Here is where the honest, contested part of the science lives. A 2021 study surveyed 185 people who had recovered from COVID-19 and found that 30 percent of them reported ongoing Long COVID symptoms. Among a subset of those subjects, 66.7 percent of the Long COVID group showed positive EA-D IgG or VCA IgM, markers consistent with EBV reactivation, compared to only 10 percent of the recovered-without-symptoms control group, a statistically significant difference.⁸ That association is genuine, measured, and interesting, and it is one reason EBV reactivation belongs in the post-viral conversation. But it is worth stating plainly what this study does and does not establish: it is one study, with a primary comparison group of 30 Long COVID subjects and 20 controls, it shows an association rather than proof that EBV reactivation causes Long COVID symptoms rather than resulting from the same underlying immune disruption, and the

researchers themselves framed their conclusion carefully, writing that Long COVID symptoms “may not be” solely a direct result of the SARS-CoV-2 virus and that they “may be” partly the result of inflammation-triggered EBV reactivation. That is appropriately hedged language, and this guide uses the same hedge deliberately rather than upgrading it into a certainty.

The broader research pattern is similarly mixed once you look past a single dramatic finding. When researchers went looking for a clear EBV signature in autoimmune thyroid disease, a condition many functional medicine sources confidently attribute to EBV, one study measuring EBV antibody levels head to head in autoimmune thyroid patients versus healthy controls found no significant difference between the two groups, while a different pathogen entirely showed a real signal in that same dataset.⁹ Contrast that with the EBV and multiple sclerosis finding from the introduction, built on a cohort of more than 10 million people with a dramatic, specific, statistically overwhelming result.¹ Both of these are real EBV-and-autoimmunity questions that real researchers have investigated. One produced a strong, specific answer. One did not. That gap is what matters. Not every “EBV causes X” claim carries the same weight of evidence, and an honest guide has to tell you which is which rather than treating every EBV association as equally proven.

What this means for your own labs. A positive VCA IgG and EBNA IgG with a normal or unremarkable EA-D most often just confirms what is true of the vast majority of adults. You were infected with EBV in the past, like almost everyone else, and your immune system has it filed away as handled. An elevated EA-D IgG, especially alongside a compatible symptom picture and in the absence of a better explanation, raises a real and reasonable hypothesis that reactivation could be contributing to how you feel, and it is a legitimate reason to investigate further rather than stop looking. What it is not, on the strength of currently available evidence, is a stand-alone diagnosis. Hold this as a hypothesis worth investigating alongside the rest of your clinical picture, not as a confirmed cause on its own.

Viral Persistence and the Reservoir Hypothesis

A hypothesis under real scrutiny. If the acute infection is over and your standard test is negative, where would ongoing symptoms even be coming from? One candidate explanation getting serious research attention is that some amount of virus, or virus-derived material, may not have fully cleared from the body the way a nasal swab or a standard blood test would detect.

The evidence, stated precisely. It is important to be precise about the state of this evidence, because it is one of the most actively investigated and least settled parts of post-viral science. The 2023 RECOVER research initiative review introduced in Key 2 lists persistent reservoirs of replicating virus or its remnants in various tissues as one of several candidate mechanisms under active investigation for Long COVID, alongside re-activation of latent pathogens such as EBV and other herpesviruses in the immune-dysregulated environment that COVID-19 creates.⁵ Researchers have reported detecting SARS-CoV-2 genetic material or viral protein in gut tissue, and in blood, of some patients months after their initial infection, in specific research cohorts, using specialized testing not available in routine clinical practice. This is a genuine and actively researched hypothesis. It is not yet a settled fact with a validated clinical test you can order to confirm it in an individual patient, and this guide is deliberately not going to overstate it as one.

The EBV reactivation picture described in Key 3 is a related but distinct form of viral persistence, and one with a much longer research history. Unlike SARS-CoV-2, EBV is a herpesvirus, and herpesviruses are built by their basic biology to persist for life in a dormant, latent state inside certain immune cells after the first infection resolves, only to reactivate periodically, usually silently and without symptoms, when the immune system's control over it weakens. A significant physical or immune stressor, and a new viral infection like SARS-CoV-2 is exactly that kind of stressor, is a biologically plausible trigger for a temporary uptick in EBV reactivation, which is the mechanism the Gold et al. findings in Key 3 are consistent with.⁸

How this guide treats an open question. Because tissue-level viral persistence is not yet something a standard clinical test can confirm or rule out in an individual, this guide treats it as a genuine hypothesis that helps explain why your immune system might still be on alert, rather than as a target you chase with an aggressive antiviral protocol on the strength of a hunch. The practical, actionable piece is supporting the immune system's own surveillance and regulatory capacity (covered in Keys 2, 5, and 9) so that whatever viral material or reactivation activity is present gets appropriately contained, rather than assuming any single supplement can be marketed as directly "killing" a persistent virus that current testing cannot even reliably locate.

The Gut and Immune Terrain

Not two separate complaints. Digestive symptoms after a viral illness, new food sensitivities, bloating, altered bowel habits, are often dismissed as unrelated to the fatigue and brain fog that brought you in, when in fact the gut and the immune system are two expressions of the same underlying terrain.

Why the gut carries a mark. Roughly 70 to 80 percent of the immune system's tissue sits in and around the gut, so it should not be surprising that a significant systemic viral illness leaves a mark there. A prospective cohort study followed COVID-19 patients from admission out to six months and compared their stool microbiome to non-COVID-19 controls. At six months, 76 percent of the COVID-19 group in this particular cohort had at least one persistent symptom (most commonly fatigue, poor memory, and hair loss), and the composition of a patient's gut microbiome at the time of hospital admission was associated with whether they went on to develop those persistent symptoms.¹⁰ Patients whose gut microbiome had returned to a profile resembling the non-COVID-19 controls by six months tended to be the ones without ongoing symptoms, while those with persistent symptoms showed lower levels of several beneficial, short-chain-fatty-acid-producing bacterial species and higher levels of a few opportunistic species. This specific cohort's 76 percent figure reflects hospitalized patients assessed for any persistent symptom at six months, a different measurement than the roughly 6 percent global Long COVID prevalence estimate cited in Key 1, which uses the stricter, population-wide WHO clinical definition. Both figures are accurate for what they each measured; they are simply not the same measurement, and conflating them would overstate the point.

The RECOVER mechanistic review from Key 2 places these gut findings inside the same broader picture, describing SARS-CoV-2's interactions with the host's microbiome and virome communities as one of the several parallel contributors to prolonged symptoms.⁵ The general logic runs in both directions, echoing what is already well established in other chronic illness contexts: a viral illness and its associated inflammation can disrupt the gut barrier and the balance of gut bacteria, and a disrupted gut barrier in turn allows bacterial fragments into circulation that keep the immune system in an activated, inflamed state, which then makes full recovery from the original viral hit slower and less complete. Each side feeds the other.

Where gut support actually fits. In post-viral recovery, supporting gut barrier integrity and microbial diversity addresses one of the specific, documented mechanisms that keeps the immune system from fully standing down, rather than serving as a generic wellness add-on. That means the same foundational levers that matter in any gut-terrain conversation: adequate fiber diversity to feed short-chain-fatty-acid-producing bacteria, identifying and removing any specific triggers your own history points to, and rebuilding barrier integrity rather than only chasing symptoms. If gut symptoms are your dominant complaint rather than a secondary piece of a post-viral picture, my gut-focused resources go deeper into that workup than this guide will; here, the point is simply that your gut is part of your immune recovery, not a separate issue running in parallel.

Mitochondrial Energy and the Post-Exertional Malaise Guardrail

Advice that can backfire. “Just push through it and build back up your stamina” is well-meaning advice from friends, family, and sometimes even well-intentioned clinicians who are more used to ordinary deconditioning than to post-viral illness. For a meaningful share of people recovering from Long COVID or a significant post-viral syndrome, that advice is not just unhelpful. It can make you measurably worse, and understanding why is essential before you design any activity plan for yourself.

What the mitochondrial research actually shows. Mitochondria are the cellular structures that convert food and oxygen into usable energy, and multiple lines of research have investigated whether post-viral fatigue involves impaired mitochondrial function. The honest state of that evidence deserves an honest summary. A systematic review of 19 studies comparing mitochondrial structure and function in ME/CFS (myalgic encephalomyelitis, chronic fatigue syndrome) patients against healthy controls found real signals of altered mitochondrial structure, genetics, respiratory function, and metabolites in multiple individual studies, but concluded that, taken as a whole, the body of research is inconsistent enough across different study designs and diagnostic criteria that it remains difficult to establish exactly what role mitochondrial change plays in the illness.¹¹ That is the fair, current answer: mitochondrial involvement is a real and active research question with supportive evidence in several individual studies, not yet a fully proven, uniform mechanism. This guide is not going to claim more certainty than the research itself currently supports, and if fatigue and cellular energy are your central concern independent of a viral trigger, my Unlock Your Energy guide covers the mitochondrial terrain in far greater depth than the scope of this guide allows.

What is much better established, and considerably more important for you to understand right now, is post-exertional malaise, often abbreviated PEM (sometimes called post-exertional symptom exacerbation, or PESE). PEM is a delayed, disproportionate worsening of symptoms, fatigue, cognitive function, pain, or all three, typically beginning hours to a day or two after physical, cognitive, or emotional exertion that would not have caused a problem before you got sick. It is one of the most frequently reported features of both Long COVID and ME/CFS, and researchers studying it have found that people experiencing PEM show a genuinely different physiological response to activity than healthy people do: reduced oxygen extraction and reduced oxidative phosphorylation capacity (the cellular energy-production process mitochondria carry out) during and after exertion, a pattern the researchers link to a combination of mitochondrial strain, microvascular dysfunction (impaired blood flow at the smallest vessel level), and ongoing low-grade immune activation working together to impair the body’s energy supply during activity.¹² This is not a motivation problem, and it is not deconditioning in the ordinary sense, even though the two can look superficially similar from the outside.

This is the single most important safety guardrail in this entire guide. Because PEM involves a genuine, measurable physiological penalty for overexertion, a standard “graded exercise therapy” approach, steadily

and linearly increasing activity on a fixed schedule regardless of symptoms, is not appropriate where PEM is present, and pushing through worsening symptoms is not the same thing as productive discomfort during ordinary exercise. A 2025 multidisciplinary compendium representing over 40 Long COVID clinical centers is explicit on this point: physical activity recommendations must be carefully tailored to a patient's current activity tolerance, because overly intense activity can trigger PEM and worsened muscle damage, and the compendium's overall management framework centers on energy conservation strategies rather than progressive exertion targets.⁴

The alternative, evidence-informed approach is called pacing, built around a concept called the energy envelope. That means identifying the amount of physical, cognitive, and emotional exertion you can currently sustain without triggering a PEM crash, and staying inside that envelope consistently, rather than alternating between big push days and crash days. Research on energy envelope maintenance in ME/CFS patients has found that people who stay within their current energy envelope tend to function better than those who consistently overextend past it, and that overextension is particularly costly for people who otherwise have more available energy to begin with, since they are the ones most tempted to push past their actual current limit.¹³ The envelope is not permanent or fixed. As the underlying immune and mitochondrial terrain genuinely improves, most people's sustainable envelope gradually widens. But it widens from consistent pacing and symptom-guided activity, not from forcing a fixed increase on a calendar regardless of how you feel the next day.

How to tell productive effort from PEM. Before increasing any activity, physical, cognitive, or social, track whether the level you are at now reliably leaves you stable the following day. If it does, small, symptom-guided increases can be reasonable. If any increase in exertion is reliably followed by a delayed crash, that is PEM, and the correct response is to reduce activity back to your stable baseline and rebuild your envelope from there with your practitioner's guidance, not to interpret the crash as a sign you need to push harder.

Autonomic Dysfunction and POTS-Type Symptoms

Not anxiety, a named syndrome. A racing heart on standing up, dizziness, lightheadedness, exercise intolerance out of proportion to any cardiac or pulmonary finding, and heat intolerance are frequently dismissed as anxiety, especially in patients whose standard cardiac workup comes back clean. For a real subset of post-viral patients, these are autonomic nervous system symptoms with an identifiable name and a legitimate, non-psychiatric physiological basis.

How the autonomic system misfires. The autonomic nervous system regulates the body's automatic functions, heart rate, blood pressure, digestion, and temperature regulation among them, without any conscious effort on your part. Postural orthostatic tachycardia syndrome, POTS, is a form of autonomic dysfunction defined by an excessive rise in heart rate upon standing, along with symptoms like lightheadedness, palpitations, fatigue, and cognitive difficulty that improve when lying back down. In 2021, the American Autonomic Society issued a formal statement addressing what it termed Long-COVID POTS, stating plainly that in the wake of COVID-19, a meaningful number of patients are developing chronic autonomic symptoms consistent with POTS, and calling for a substantial investment of clinical and research resources to properly understand and manage it.¹⁴ This is a real, medically recognized phenomenon, formally acknowledged by the specialty society devoted to autonomic disorders, not an alternative-medicine invention.

POTS and related dysautonomia are not unique to COVID-19. The same pattern has long been described following other viral illnesses, including EBV, and the RECOVER mechanistic review discussed earlier explicitly draws the comparison between Long COVID and other virus-associated chronic conditions, noting dysfunctional brainstem and vagus nerve signaling and autonomic dysfunction as a shared mechanism across post-viral syndromes generally, not a COVID-specific quirk.⁵ The proposed mechanisms include direct viral or immune-mediated injury to the autonomic nerves themselves, deconditioning from a period of reduced activity (which can worsen an existing autonomic problem even when it is not the original cause), and the same autoimmune molecular mimicry mechanism described in Key 2, in which antibodies generated against a viral protein cross-react with the body's own autonomic nerve receptors.

Practical starting points, not a diagnosis. If standing up reliably produces a racing heart, lightheadedness, or a rush of symptoms that improve when you lie down, that pattern is worth bringing to your practitioner by name, POTS or orthostatic intolerance, rather than letting it get filed under general anxiety. Practical, low-risk starting points that a practitioner may recommend include increasing fluid and electrolyte intake, compression garments, and avoiding prolonged standing, all of which are supportive measures to discuss with your care team rather than a substitute for a proper autonomic evaluation, especially since some POTS treatments involve prescription medications that need to be managed by your physician directly.

Mast Cells and Histamine, and Why Everything Feels Reactive

Why everything suddenly feels reactive. New food reactions, unpredictable hives or flushing, worsening seasonal allergies, and a sense that your body has become reactive to things that never used to bother you are a distinct and under-recognized piece of the post-viral picture, separate from the fatigue and brain fog that usually bring people in the door first.

How mast cells drive the pattern. Mast cells are immune cells that release histamine and a range of other inflammatory mediators as part of a normal allergic and immune response. Mast cell activation syndrome, MCAS, describes a state in which mast cells release these mediators too readily or too often, producing a wide, seemingly unrelated collection of symptoms that includes flushing, hives, gastrointestinal upset, headaches, and rapid heartbeat. A 2021 study compared self-reported symptoms in 136 people with Long COVID, 136 general population controls, and 80 patients formally diagnosed with MCAS, and found that the Long COVID group reported mast cell activation symptoms at a severity that closely matched the established MCAS patient group, and at a rate clearly elevated above the general population controls.¹⁵ The researchers proposed that aberrant mast cell activation triggered by SARS-CoV-2 infection, through several possible mechanisms, may underlie part of the symptom picture in Long COVID, which the authors framed as a promising and specific lead toward more effective treatment, appropriately, as a hypothesis supported by this comparison rather than a fully proven mechanism.

This overlaps meaningfully with the immune activation picture from Key 2 and the gut terrain from Key 5. A gut barrier that has become more permeable following a significant viral illness allows more triggers through to interact with the substantial population of mast cells that live in the gut lining itself, and a nervous system running in a heightened sympathetic state (relevant to the autonomic symptoms in Key 7) can itself lower the threshold at which mast cells degranulate and release their mediators. None of these systems operate in isolation from each other.

Where to start if this is you. If new or worsened reactivity, whether to foods, fragrances, alcohol, or temperature changes, developed alongside your other post-viral symptoms, mention it specifically to your practitioner rather than treating it as a separate, unrelated allergy question. A low-histamine trial period, addressing gut barrier integrity as covered in Key 5, and identifying and removing specific personal triggers are the typical starting points, well before considering any mast-cell-stabilizing medication, which should be a conversation with your prescriber given the overlap with other medications you may already be taking.

What Appropriate Functional Testing Actually Looks Like

The question worth asking first. Given everything covered so far, a reasonable next question is what you should actually ask a practitioner to test. The honest answer is that no single test confirms this picture, which is exactly why a scattershot approach to lab ordering wastes money without producing clarity.

What the current guidance actually says. As the 2025 Long COVID compendium states plainly, there is currently no single laboratory finding that definitively confirms or rules out Long COVID, so a basic laboratory assessment is recommended for essentially everyone, with additional, more targeted testing guided by each patient's specific symptom pattern rather than ordered as a blanket panel for every patient regardless of presentation.⁴ Translating that principle into practice, a useful post-viral workup is generally built in layers, starting broad and narrowing based on what the first layer shows, rather than ordering every specialty panel at once.

A sensible first layer looks at the magnitude of general immune and inflammatory activity. That means a complete blood count with differential, inflammatory markers, a comprehensive metabolic panel, and a thyroid panel, since thyroid dysfunction can produce a fatigue picture that overlaps with post-viral illness and is far cheaper and faster to rule in or out first. From there, targeted testing depends on which of this guide's chapters your own symptom pattern points toward. If EBV reactivation is a live question given your history and symptoms, a full EBV panel, VCA IgM, VCA IgG, EBNA IgG, and EA-D IgG together, read as a pattern rather than any single marker in isolation, is far more informative than a single EBV antibody result, exactly per the honesty section in Key 3. If gut symptoms are prominent, a comprehensive stool or gut microbiome assessment is a reasonable next step, per Key 5. If autonomic symptoms like the ones in Key 7 are prominent, a simple in-office orthostatic vital signs check, heart rate and blood pressure lying down and then standing at intervals, is a low-cost, high-value screening step before more specialized autonomic testing. Immune panels that look at broader immune regulation, including markers of autoimmune activity and immune signaling, can be useful when the symptom picture suggests immune dysregulation extending beyond simple post-viral fatigue, particularly given the real, documented example from Key 2 of antibodies against the body's own antiviral signaling machinery.⁶ And for the mitochondrial and energy piece, organic acids testing can characterize markers of cellular energy metabolism, though as Key 6 covers honestly, the research connecting any single one of these markers definitively back to post-viral illness specifically is still developing, so this piece of testing is best understood as characterizing your terrain broadly rather than confirming a specific mechanism.

Building the workup in layers. Bring your practitioner a clear timeline (when you were infected, what came first, what has changed since), and let your own symptom pattern, not a fixed template, drive which of these layers gets ordered. A workup built this way costs less, produces more actionable information, and avoids the trap of chasing a single abnormal number that may not actually explain how you feel.

Red Flags: When to Seek Immediate Care

Post-viral recovery is rarely a medical emergency, but certain symptoms always warrant prompt conventional evaluation rather than a supplement or a wait-and-see approach, no exceptions. Seek emergency or urgent medical care immediately for:

- Chest pain or pressure, especially with exertion
- Shortness of breath at rest, or a marked new inability to catch your breath
- Fainting or a real loss of consciousness (as opposed to lightheadedness that resolves on sitting or lying down)
- New weakness, numbness, or difficulty speaking on one side of the body
- Symptoms of a blood clot, such as one-sided leg swelling, warmth, or pain, or sudden severe shortness of breath
- A rapid, unexplained worsening of any symptom, rather than the slower ebb and flow typical of post-viral illness

This list is not decorative. Large healthcare-database research following COVID-19 survivors for a full year found a meaningfully increased risk, beyond the first 30 days after infection, of a range of cardiovascular problems, including blood clots, heart rhythm disturbances, heart attacks, and heart failure, and this increased risk was present even in people who were never hospitalized for their original infection.¹⁶ That is precisely why the red flags above are not optional caution language. They reflect a real, measured, elevated risk in exactly the population this guide is written for.

Your Staged Recovery Roadmap

Post-viral recovery is rarely linear, and the timeline below reflects my general clinical experience at Root Health with the pace of recovery, not a guarantee for any individual case. Your own timeline depends on how long you have been unwell, which mechanisms from this guide are most active in your case, and how consistently pacing and terrain support are followed.

Weeks 1 to 4: Stabilize and map the terrain. Focus on identifying your current energy envelope (Key 6) without trying to expand it yet, basic labs to rule out other explanations (Key 9), and beginning gut barrier and nutrient foundation support (Key 5). Most people notice their first small, real improvements in this window, typically in sleep quality or the predictability of their symptoms, before they notice a big jump in energy.

Months 2 to 3: Targeted layer. With your first labs back, this is when EBV-specific findings, autonomic testing, or mast cell patterns get addressed directly rather than generically. Energy and symptom improvement typically becomes more noticeable here, though pacing inside your envelope remains the priority over pushing activity.

Months 3 to 6: Consolidation. This is the phase where the terrain work you started earlier either holds or does not, and where the picture separates. In my clinical experience, a straightforward post-viral or EBV-reactivation picture consolidates faster than a complex, longer-standing, multi-system one. I am deliberately not going to hand you a number of months, because I do not have trial data on recovery timing in this population and my own case experience is not a substitute for it. What I can tell you is what to measure: compare your current worksheet against your own baseline from week one, and watch whether your activity ceiling is rising and whether a crash still reliably follows exertion. That comparison is a far better read on your progress than any calendar I could give you. Pacing consistency is the biggest single lever either way.

Beyond 6 months: For people whose picture is more complex or longer-standing, the work continues but more gradually, and this is where a structured, practitioner-guided approach earns its value most, since it is exactly the point at which self-directed trial and error tends to stall out.

Your Post-Viral Recovery Worksheet

Use this each week, ideally at the same time and day, to build the pattern recognition that makes pacing and practitioner conversations both far more productive.

Part 1: Energy Envelope Check

Day	Physical exertion today (low / moderate / high)	Cognitive exertion (low / moderate / high)	Symptoms the next day (same / better / worse)	Notes
Mon				
Tue				
Wed				
Thu				
Fri				
Sat				
Sun				

If “worse” shows up the day after a “high” exertion day more than once, that is your current envelope ceiling. Note it, and treat it as real data rather than a limitation to push through.

Part 2: Symptom Tracker

Once daily, rate fatigue, brain fog, heart racing on standing, digestive symptoms, new food or scent reactivity, and sleep quality, each from 0 (absent) to 4 (severe). Sleep is reverse-scored, so 0 is excellent sleep and 4 is very poor. Trends across two to four weeks are far more useful to a practitioner than any single day's numbers.

Part 3: Before Your Next Appointment

- When you first noticed these symptoms, and what infection or illness came before them
- The single symptom most limiting your daily function right now
- Any reliable pattern you have noticed between an activity and a delayed crash (PEM, Key 6)
- Any new reactivity to food, scent, or medication since you got sick
- Any red flag symptom from the list above, even once

Take the Next Step: Your Free Root Discovery Session

Everything in this guide is meant to help you understand your own picture and walk into a clinical conversation with better questions, not to replace that conversation. The Root Discovery Session is a complimentary consultation with my team to review your specific timeline, symptoms, and prior testing, and to talk through whether a structured, lab-directed functional medicine approach is the right next step for you. There is no obligation, and if you decide to move forward, this conversation is simply the first step.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Not ready for a conversation yet? Start tracking.

The recovery worksheet earlier in this guide works on paper. The digital version does the same job on your phone, keeps your history in one place, and shows you the pattern that matters most here, which is whether a crash reliably follows exertion on a delay and where your true activity ceiling sits. Two weeks of honest data makes any future appointment, with our team or your own physician, dramatically more useful than memory alone.

If you would rather start on your own, **start your free tracker at rootcauseunlocked.com/tracker.**

Seay Wellness | 503 US-70 East, Suite L, Garner, NC 27529
rootcauseunlocked.com

This guide is provided for general educational purposes and does not constitute individualized medical advice, diagnosis, or treatment. It does not diagnose Long COVID, chronic EBV, POTS, mast cell activation syndrome, or any other condition. Always consult your physician before starting any new supplement or exercise program, particularly if you are pregnant, nursing, or taking prescription medications, and before making any change to a prescribed medication. If you experience any of the red flag symptoms listed in this guide, seek emergency or urgent medical care immediately rather than waiting to discuss them at a future appointment.

References

Every citation below was independently retrieved and confirmed against its live PubMed record (research plus summary and/or fetch for the full abstract) before use. None are drawn on trust from any prior document. Full verification detail, including claims investigated and left uncited, is in the companion file Post-Viral-Recovery-Citation-Ledger.md.

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From Understanding to Doing

Everything up to this point was about understanding what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek emergency care now for chest pain or pressure, new or worsening breathlessness, coughing blood, a swollen or painful calf, fainting rather than lightheadedness, new weakness or numbness on one side, trouble speaking, or severe headache of sudden onset. Blood clots are a recognized complication after COVID, and post-viral fatigue is never the safe assumption when these appear.

Also see a doctor promptly for unintentional weight loss, fever or night sweats that persist, a new lump, or fatigue that has worsened steadily rather than fluctuating.

If you have thoughts of harming yourself, call or text 988 in the United States, or go to an emergency department.

Read this before anyone prescribes you exercise

If your symptoms reliably worsen a day or two after exertion, and that worsening is out of proportion to what you did, that is **post-exertional malaise**. It is Key 6 of *Unlock Your Post-Viral Recovery* and it is the guardrail on everything else.

Where post-exertional malaise is present, graded exercise is not the answer and pushing through causes harm. This matters more in this kit than anywhere else on the shelf, because you will find exercise reconditioning recommended for one of the other things in this guide, and the two pieces of advice collide. I deal with that collision head-on in the autonomic section rather than leaving you to discover it.

The nine keys, and what tests each one

Unlock Your Post-Viral Recovery works through nine root causes. Blood work reaches some of them and not others, and being clear about which is the point of this guide.

Key	What it means	How you find it
1. Persistence	What “the infection never fully left” does and does not mean	Mostly a research question. No routine test
2. Immune activation	The inflammatory signature	hs-CRP, CBC with differential
3. EBV serology	What the antibody panel actually says	VCA IgG, VCA IgM, EBNA-1 IgG. See below, this is the most misread panel in the category
4. Reservoir	The viral reservoir hypothesis	Research setting. Not a test you can order
5. Gut and immune terrain	Absorption and the microbial contribution	Celiac serology, stool analysis where symptoms justify it
6. Mitochondria and PEM	Energy production, and the post-exertional guardrail	History, not a test. The most consequential finding here
7. Autonomic	POTS-type symptoms, orthostatic intolerance	A standing test in the office. Not a blood test
8. Mast cells and histamine	Why everything feels reactive	Baseline tryptase, with strict criteria. See below
9. Testing	Choosing the right panel rather than the biggest one	This guide

Two of the nine cannot be tested outside research, two more are answered by history or a bedside test rather than blood, and one, EBV serology, is more often misread than read. That is the honest map.

Key 3: EBV serology, the most misinterpreted panel in this category

This is the section I would keep if I could keep only one, because a great deal of money is spent on the back of a misreading.

What the panel measures. The standard EBV panel reports antibodies to viral capsid antigen, both IgM and IgG, and to Epstein-Barr nuclear antigen 1. In an evaluation of these assays for determining EBV serostatus, patients without heterophile antibodies who had **VCA IgG and EBNA IgG antibodies were supporting past infection** (PMID 28661055).

What that means for you, stated plainly.

- **VCA IgG positive plus EBNA-1 IgG positive is the signature of past infection.** It means the virus met your immune system at some point, most often in adolescence.
- **Roughly nine in ten adults carry that pattern.** It is close to universal.
- **It is therefore not a diagnosis, and it does not explain your fatigue.** If a positive EBV IgG were sufficient to explain post-viral symptoms, most of the adult population would have them.

Where the misreading happens. A panel comes back with “high” VCA IgG, the number is flagged above a reference line, and it gets presented as proof of chronic reactivated EBV driving the illness. Titre height on IgG is not a reliable measure of current viral activity, and a raised IgG in someone with past infection is an ordinary finding.

What would actually be different. Chronic active EBV disease is a genuinely recognized entity, and it is **rare and severe**, presenting with persistent fever, organ involvement, and lymphoproliferative complications rather than with fatigue alone. It is diagnosed in specialist settings, usually with viral load quantification rather than antibody titres. If a clinician genuinely suspects it, that is a haematology referral, not a supplement protocol.

So what should you do with an EBV panel? Run it once if it has never been done, mainly to establish serostatus. Do not repeat it to track progress, because it will not track anything useful, and do not let a positive IgG become the explanation that stops the search.

Key 7: the autonomic system, and the contradiction you need to know about

This is the key most often missed, and it is also where the safety advice in this field genuinely conflicts.

How common it is. A cross-sectional study of long COVID patients found dysautonomia common and contributing to overall symptom burden, concluding that identifying it has diagnostic and therapeutic implications (PMID 41951431). A systematic review examined the prevalence of orthostatic intolerance, postural orthostatic tachycardia syndrome, and orthostatic hypotension in post-acute sequelae of COVID (PMID 41584279).

What it feels like. Lightheaded on standing, heart racing when upright, worse in heat, worse after meals, better lying down, brain fog that lifts when horizontal. If you have quietly started sitting down in showers, that is this.

How it gets found, and it is not a blood test. A standing test in the office. You lie down for ten minutes while heart rate and blood pressure are recorded, then stand while they are recorded at intervals over ten minutes. A sustained heart rate rise of 30 beats per minute or more in adults on standing, without a fall in blood pressure, is the POTS pattern. Ask for this by name. It takes fifteen minutes and costs nothing.

The contradiction, stated directly

For POTS, the literature recommends exercise reconditioning. Cardiovascular deconditioning, meaning cardiac atrophy and low blood volume, contributes significantly to POTS and its disability, so physical reconditioning with exercise training is a recommended approach, alongside measures such as increased salt and fluid intake and compression, while the authors note that randomized trials are still needed to evaluate these non-pharmacological treatments (PMID 30001836).

For post-exertional malaise, exercise is contraindicated. Pushing causes harm.

Many people in this population have both. So which wins?

The post-exertional malaise rule wins. If you have PEM, an exercise prescription designed for POTS can make you worse, and the reconditioning literature was not written with PEM in mind. This is the single most important thing in this guide to raise with a clinician, because a well-meaning graded exercise plan aimed at your POTS is a realistic way to lose six months.

What is safe to do meanwhile, and is recommended in the same non-pharmacological literature: increase salt and fluid intake if your clinician agrees it is appropriate for you, use compression garments, change position slowly, and use recumbent or horizontal movement rather than upright exertion if you are attempting any activity at all. Those address the orthostatic side without triggering the PEM side.

Key 8: mast cells and histamine, with the criteria that actually apply

Why it is here. A proportion of people in this group describe reacting to foods, smells, medications, heat, and alcohol in ways they never did before.

What has been reported. A study of long COVID patients with cardiovascular manifestations attributed to mast cell activation reported that antihistamines improved cardiovascular manifestations and other symptoms, with no significant change in controls, and the authors take this as supporting a role for mast cell activation in long COVID pathophysiology (PMID 37529714).

Read that carefully. It is a small non-randomized study, the comparison group was not a placebo arm, and “responded to antihistamines” is suggestive rather than diagnostic. It is enough to make a trial of an over-the-counter antihistamine a reasonable conversation with your clinician. It is not enough to establish a diagnosis.

And the diagnostic discipline matters here. Mast cell activation syndrome has consensus criteria requiring typical episodic symptoms in at least two organ systems, a substantial rise in serum tryptase above the person's own baseline during an event, and response to mast-cell-directed treatment. A critical review found that the alternative criteria circulating in practice often lack specificity and validation, and that an increasing number of patients are being given a misdiagnosis on the strength of them (PMID 34166845).

Practically: a baseline serum tryptase is reasonable if reactions are systemic. A tryptase drawn during an actual event, compared against that baseline, is the informative test. The Allergy kit covers this in more depth, including hereditary alpha-

tryptasemia, which explains most raised baseline tryptase results.

Keys 2 and 5: the terrain, and the ordinary blood work

Inflammation. hs-CRP and a CBC with differential. Do not expect them to be dramatic; in most post-viral presentations they are unremarkable, and a normal CRP does not mean nothing is wrong.

The gut. A case-based systematic review of brain-gut axis approaches in long COVID and ME/CFS notes that the two conditions share persistent fatigue, post-exertional malaise, and gastrointestinal dysfunction, with growing evidence implicating brain-gut axis dysregulation, while calling for larger mechanistic studies and rigorously designed randomized trials (PMID 41668172). Directional, and honest about its own limits.

Practically: celiac serology while still eating gluten, and stool analysis only if gut symptoms genuinely justify it.

The deficiencies worth excluding, because they are common, correctable, and produce exactly these symptoms. They are ferritin with a full iron panel, B12, RBC folate, homocysteine, vitamin D, and thyroid function with TSH, free T4 and free T3. Ferritin is the one most often waved off, because a result inside a lab's standard range still gets reported as normal. Read it against the functional window in the table below rather than against the lab's flag, and see the supplement guide in this kit for the trial evidence behind that threshold.

How common is this, and is it real

Both questions get asked. A systematic review and meta-analysis of prospective cohort studies found long COVID affects over one-fifth of people who had SARS-CoV-2, with fatigue and breathlessness persisting in many, and female sex and severe acute infection increasing risk (PMID 41462874). A separate meta-analysis of observational studies emphasizes that reported prevalence varies substantially with definitions and that results should be read as a descriptive summary of heterogeneous evidence rather than a single precise rate, while calling for rehabilitation pathways and multidisciplinary care (PMID 42428921).

Long COVID exists and is common, and the prevalence numbers are genuinely uncertain because the definitions vary.

Functional ranges for the markers in this guide

Clinical targets used at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose.

Marker	Units	Functional target	Approximate standard range
Ferritin	ng/mL	50 to 150	25 to 300
Serum iron	ug/dL	65 to 110	40 to 150
TIBC	ug/dL	290 to 370	250 to 400
Transferrin saturation	%	20 to 40	15 to 45
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Homocysteine	umol/L	5 to 7	under 10.0
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
hs-CRP	mg/L	under 0.5	under 1.0
White blood cells	K/uL	5.0 to 7.5	3.5 to 10.5
Lymphocytes	%	25 to 40	15 to 45
Glucose, fasting	mg/dL	75 to 86	70 to 99
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0

On homocysteine, lower is not better. The 5 to 7 target is a window, not a floor. A homocysteine driven below it points toward overmethylation and B vitamin intolerance, which is its own problem rather than a better result. Read an elevated result as a prompt to find the short input, B12, folate, B6, thyroid, or kidney function, not as a number to push down as far as it will go.

Thyroid targets are deliberately not published here. I only publish ranges I can transcribe exactly from a validated source, and the right thyroid target depends on age, pregnancy, and whether you are already on replacement.

The conversation with your doctor

A request list you can bring

- CBC with differential, hs-CRP
- Ferritin, serum iron, TIBC, transferrin saturation
- Vitamin B12, RBC folate, homocysteine
- 25-hydroxy vitamin D
- TSH with free T4 and free T3
- Celiac serology, drawn while still eating gluten
- EBV panel **once**, to establish serostatus, if never done
- Baseline serum tryptase, if reactions are systemic or involve more than one organ system
- **A ten-minute standing test**, which is free and is the highest-yield item on this list

The sentence that changes the appointment

“My symptoms are worse when I am upright and better lying down. Could we do a ten-minute standing test with heart rate and blood pressure?” That is specific, cheap, and it finds the thing most often missed.

And the sentence that protects you: “My symptoms reliably get worse a day or two after exertion, so I need any activity plan to account for post-exertional malaise.”

What these labs cannot tell you

- **Whether a virus is still active in your tissue.** The reservoir question is a research question and there is no clinical test.
- **Whether your EBV is reactivated.** A positive IgG means past infection, which is true of most adults.
- **Whether you have post-exertional malaise.** History answers that, and it matters more than the panel.
- **How long recovery will take.** Nobody can tell you that honestly, and anyone who does is guessing.

When to seek clinical review

Emergency. Chest pain, new or worsening breathlessness, coughing blood, swollen painful calf, fainting, one-sided weakness or numbness, trouble speaking, sudden severe headache.

Soon. Unintentional weight loss, persistent fever or night sweats, a new lump, steadily worsening rather than fluctuating symptoms, fainting on standing, or a resting heart rate that has changed markedly.

Now: thoughts of harming yourself. Call or text 988.

Your next step

The companion guides do the rest. **The Post-Viral Diet Guide** covers the gut, histamine, and eating within an energy budget, and the supplement guide covers pacing, what has been studied, and what to skip.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, start tracking, and track exertion alongside symptoms so the delay between them becomes visible:

Free symptom tracker: rootcauseunlocked.com/tracker

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Dr. Adam Seay, DC · Seay Wellness / Root Health, Garner, NC
rootcauseunlocked.com · (919) 662-0520

Educational content only. Not medical advice, not a diagnosis, and not a substitute for your clinician.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek emergency care for chest pain or pressure, new or worsening breathlessness, coughing blood, a swollen painful calf, fainting rather than lightheadedness, one-sided weakness or numbness, trouble speaking, or sudden severe headache.

If you are pregnant, nursing, underweight, diabetic, or have a history of disordered eating, talk with a clinician before restricting anything. In this population an unnecessary restriction costs more than usual, for the reason in the next section.

If you have thoughts of harming yourself, call or text 988.

The constraint that governs this whole guide

Cooking is an activity. Shopping is an activity. Standing at a counter chopping vegetables comes out of the same daily budget as showering, working, and answering messages.

Most post-viral diet advice is written as though the reader has normal energy to implement it, then prescribes an elimination protocol with three cooked meals a day. The person following it spends their entire envelope on food preparation and crashes.

So every strategy here has to cost less energy than it returns, and where something is expensive to run, I say so.

If your symptoms reliably worsen a day or two after exertion, that is post-exertional malaise, and this is not a nicety. Read the pacing section of the supplement guide first.

Where food sits among the nine keys

Unlock Your Post-Viral Recovery works through nine root causes. Food reaches three of them, which are the gut and immune terrain, the histamine and mast cell picture, and the nutrient side of the mitochondrial key. It reaches neither viral persistence nor post-exertional malaise, and it does not fix orthostatic intolerance, though the salt and fluid part below helps that one.

Knowing which keys food can actually touch is what stops you spending six months on the wrong one. The lab guide maps all nine.

Part one: the two changes that help most people here

Salt and fluid, if standing is a problem

This is the highest-yield dietary change in this guide for anyone whose symptoms are worse upright, and it is unusual in being a dietary change with a physiological target rather than a general health goal.

Increased salt and fluid intake sits among the non-pharmacological measures described for postural orthostatic tachycardia syndrome, alongside compression, with the authors noting that randomized trials are still needed to evaluate these approaches (PMID 30001836).

How to do it. Fluid steadily across the day rather than in large boluses. Salt added deliberately to food, or an electrolyte drink without much sugar. Drink a glass of water before standing up in the morning, before showers, and before anything that reliably makes you lightheaded.

Clear this one with a clinician first. Salt loading is genuinely inappropriate in high blood pressure, heart failure, and kidney disease, and this is the one recommendation in the kit where getting it wrong for your situation matters.

Eating in a way that does not itself cause symptoms

Large meals pull blood to the gut and reliably worsen both orthostatic symptoms and post-meal crashes in this population.

- **Smaller meals, more often.** Four or five small ones beats two or three large ones here.
- **Protein at breakfast**, which steadies the rest of the day.
- **Sit down to eat**, and stay sitting for a while afterwards if standing after meals is a problem.
- **Watch the post-meal window.** If you reliably feel worse forty minutes after eating, meal size is the first thing to change, before the food list.

Part two: the histamine question, scoped honestly

Key 8 of the ebook covers mast cells and histamine, and a share of readers will recognize themselves in it, reacting to foods, alcohol, heat, and smells in ways that are new since the illness.

What has been reported. In long COVID patients with cardiovascular manifestations attributed to mast cell activation, antihistamines improved cardiovascular and other symptoms, with no significant change in controls (PMID 37529714). That is a small non-randomized study without a placebo arm, and it is enough to make the possibility worth taking seriously rather than enough to diagnose anything.

What a dietary trial looks like, and what it costs. A low-histamine trial is a genuine option and it is one of the more expensive things in this guide in energy terms, because the operating principle is freshness rather than a forbidden list. Meat cooked and eaten the same day is low histamine; the same meal on day three is not. That means more frequent cooking, and energy for cooking is exactly what you are short of.

So run it as a bounded, cheap version:

- **Four weeks, not indefinitely.**
- **Change the freshness variable first**, before you remove whole food groups. Cook, portion, and freeze immediately rather than refrigerating leftovers for days. For many people this is most of the available benefit and it costs no extra cooking.
- **Then remove the top few candidates** if that alone was not enough, meaning aged cheese, cured and smoked meat, fermented foods, alcohol, and leftovers of any protein.
- **Reintroduce one at a time afterwards.** Skipping this is how a four-week trial becomes a permanently narrowed diet.
- **If four weeks changes nothing, stop.** Do not escalate to a stricter version.

The Allergy Diet Guide covers the histamine evidence in full, including the placebo-controlled challenge data, and it is worth reading before you commit to this.

Part three: the gut key, without overclaiming

Why it matters concretely. Absorption. You cannot correct an iron or B12 deficiency through a gut that is not absorbing, and those deficiencies produce exactly these symptoms.

Get celiac testing before removing gluten. The serology is only interpretable while you are still eating it. Remove it first and you lose a clean answer without a deliberate gluten challenge later.

What the research supports, and its limits. A case-based systematic review of brain-gut axis approaches in long COVID and ME/CFS notes the shared features of persistent fatigue, post-exertional malaise, and gastrointestinal dysfunction, with growing evidence implicating brain-gut axis dysregulation, while calling for larger mechanistic studies and rigorously designed randomized trials (PMID 41668172). A review of long COVID management discusses probiotics for gastrointestinal health and wellbeing and concludes that large-scale randomized controlled trials are warranted to validate their efficacy (PMID 41862909).

Read both as directional. Plausible, actively researched, not demonstrated. I am not going to tell you a gut protocol resolves post-viral illness, because no trial says so and in this population the cost of a wrong bet is measured in months.

Part four: eating within an energy budget

This is the practical heart of the guide.

Cook once, eat three times. Batch on your best day, portion immediately, freeze. Frozen portions, not five-day-old fridge leftovers, which matters doubly if histamine is part of your picture.

Keep a stocked no-cook tier at all times. Treat it as a normal part of the system rather than a failure plan. Stock tinned fish, pre-cooked rice, frozen vegetables, eggs, nut butter, fruit, and plain yogurt if dairy is fine for you.

Lower the bar for what counts as a meal. Protein plus a vegetable plus a carbohydrate. It does not need assembling or photographing.

Convert money into energy where you can. Pre-chopped vegetables, frozen produce, rotisserie chicken, grocery delivery. In this condition that is the correct direction of trade.

Sit down. Every time. Standing to eat costs more than you think and it worsens post-meal orthostatic symptoms.

A rhythm rather than a menu. Batch two proteins and two vegetables on your best day, cook a grain that keeps, rotate for three days, use the no-cook tier for two, repeat.

What I deliberately left out, and why

- **Elimination diets beyond the bounded histamine trial.** Expensive to run in energy terms, they narrow nutrition, and the yield is low. Celiac testing is the exception and it is a blood test, not a diet.
- **Extended fasting.** Skipping meals works against the blood sugar and orthostatic work above, and in this population it commonly backfires.
- **Antiviral diets aimed at “chronic EBV.”** They rest on a misreading of the serology, which the lab guide covers. Roughly nine in ten adults carry the antibody pattern being treated (PMID 28661055).
- **Detox and cleanse protocols.** No outcome evidence, real energy cost.
- **Any plan requiring three cooked meals a day.** If a plan assumes energy you do not have, it is not a plan for you.

Telling you what to skip is part of what you paid for, and here it is also giving you your afternoon back.

Running a fair experiment

Weeks 1 and 2, baseline. Change nothing. Score energy out of ten, mark crash days, and note what you did the day before each crash.

Weeks 3 to 6, one change. Salt and fluid if you are orthostatic. Meal size if you crash after eating. The freshness rule if histamine is your thread.

Week 7, judge honestly. Compare against baseline. Fewer crashes is a better signal than more energy, and it moves first.

Then one more change. Never two at once.

Realistic expectations

What is realistic. Better tolerance of being upright within a couple of weeks of getting salt and fluid right. Fewer post-meal crashes from changing meal size. Genuine improvement over months if a real deficiency gets corrected.

What is not realistic. Diet resolving post-viral illness. The recent literature calls for rehabilitation pathways and multidisciplinary care precisely because no single intervention does this (PMID 42428921).

And the thing worth saying plainly. If you have been told this would resolve if you just ate better, and you have been eating well for a year, the problem is not your diet and it is not your discipline.

When to seek professional care

Emergency. Chest pain, new or worsening breathlessness, coughing blood, swollen painful calf, fainting, one-sided weakness, trouble speaking, sudden severe headache.

Soon. Unintentional weight loss, persistent fever or night sweats, steadily worsening rather than fluctuating symptoms, fainting on standing, a diet narrowed below roughly twenty foods, or a markedly changed resting heart rate.

Now: thoughts of harming yourself. Call or text 988.

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your labs, your history, and whether post-exertional malaise and orthostatic intolerance are part of your picture.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, track first, and track exertion alongside symptoms so the one-to-two-day delay becomes visible:

Free symptom tracker: rootcauseunlocked.com/tracker

Read the companion documents alongside this one. **Understanding Your Post-Viral Labs** covers what EBV serology really means and the standing test worth asking for, and **The Post-Viral Supplement Guide** covers pacing and what has actually been studied.

References

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If you have thoughts of harming yourself, call or text 988 in the United States.

In the US, supplements are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed, with some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions. It names the funding and the evidence tier, and lets you see both.

The guardrail, before anything else

If your symptoms reliably worsen a day or two after exertion, do not let anyone talk you into pushing through. That is post-exertional malaise, it is Key 6 of the ebook, and where it is present graded exercise causes harm.

This matters more here than anywhere else on this shelf, because **you will be told to exercise for your POTS.** The lab guide explains that collision in full. The short version is that if you have post-exertional malaise, the PEM rule wins, and any activity plan has to be built around it.

Two foundations that outrank everything you can buy

Pacing

A scoping review of pacing in myalgic encephalomyelitis and chronic fatigue syndrome describes it as a prominent energy-management strategy while stating plainly that the literature base is not yet sufficient to dictate treatment practice (PMID 37838675). Co-production work with adults living with ME/CFS places energy management at the center of daily life alongside sleep, diet, and psychological support (PMID 39961545). Post-infectious conditions are increasingly recognized as a group needing better research and inclusive trial design (PMID 41967005).

The honest read: pacing is the consensus practical approach, its trial base is thinner than it should be, and it is the approach least likely to harm you. In a condition where the standard alternative caused harm, that last point carries real weight.

In practice:

- Plan to your **average** day, not your best one.
- Short blocks with rest between, rather than one push and a crash.
- Stop before you are empty. Finishing the task is what costs you tomorrow.
- Track exertion alongside symptoms, so the one-to-two-day delay becomes visible.

The orthostatic basics, if standing is part of your problem

If you are lightheaded upright, racing when you stand, and better lying down, these are the non-pharmacological measures described alongside exercise reconditioning in the POTS literature, and unlike the exercise part they do not conflict with pacing: increased salt and fluid intake where your clinician agrees it is appropriate, compression garments, and slow position changes (PMID 30001836). The same source notes that randomized trials are still needed to evaluate these non-pharmacological treatments, so take them as sensible and not as proven.

Salt is the one to clear with a clinician first, because it is not appropriate in high blood pressure, heart failure, or kidney disease.

Sleep, which is unrefreshing here rather than merely short

Unrefreshing sleep is a defining feature of this illness, not a side effect of it. People in this group commonly sleep eight or nine hours and wake as though they had not slept, and that is different from insomnia.

What that changes. Chasing more hours does not fix unrefreshing sleep. What helps is protecting the regularity, because an erratic schedule makes an already-broken repair process worse. That means a fixed wake time, light in the morning, and a wind-down that starts before you are desperate.

And where insomnia is genuinely the problem too, the treatment has real evidence. Cognitive behavioural therapy for insomnia produces effects on depressive symptoms sustained across the year following treatment in a systematic review and meta-analysis (PMID 42435604), and fully automated digital delivery has its own meta-analysis as well (PMID

42240717). The digital versions matter here specifically, because attending weekly in-person appointments is itself an exertion this population may not have.

Rule out sleep apnea if you snore, wake with headaches, or someone has seen you stop breathing. It is common, treatable, and it will not respond to anything in this guide.

The stress axis, stated carefully

There is a genuine finding here, and it is not the one usually sold.

A 2026 meta-analysis examining the neuroendocrine signature of myalgic encephalomyelitis and chronic fatigue syndrome found evidence for a bioactive cortisol deficit alongside exaggerated feedback sensitivity, integrating neuroendocrine and psychological perspectives to describe chronic stress maladaptation in fatigue syndromes (PMID 42026257).

Read the difference. That is not “your adrenals are exhausted,” and it does not mean an adrenal support product will fix it. It is a subtle regulatory finding in a diagnosed population, from a meta-analysis, and what follows from it practically is unglamorous: the stress axis is genuinely involved, so load management is part of treatment rather than a lifestyle suggestion. I do use four-point diurnal cortisol testing clinically, for the shape of the curve rather than to diagnose a named adrenal condition, which is a distinction worth holding onto in a market that blurs it.

In practice, and none of it costs anything: five to ten minutes daily of slow breathing with a longer exhale than inhale, a protected sleep window, and pacing. Being disbelieved for months is its own load, and naming that with someone competent is part of the work rather than separate from it.

How to read this guide

Test before you supplement, wherever a test exists. In this population, correcting a measured deficiency is the intervention most likely to do something.

Give it eight to twelve weeks, and track energy out of ten plus crash days.

One change at a time. Symptoms fluctuate on their own here more than in almost any condition, and a stack teaches you nothing.

Start low. Sensitivity to supplements is common in this group, and the usual starting dose is often too much. Half doses for the first week are reasonable.

Tier 1: correct what is measurably low

The lab guide covers what to test. The principle is the one I apply to every fatigue picture. Iron, B12, folate, vitamin D, and thyroid status are common, correctable, and produce exactly these symptoms.

Iron deserves the specific mention. In a double blind randomised placebo controlled trial of non-anaemic women with unexplained fatigue, iron improved fatigue and the benefit was confined to those with **ferritin at or below 50** (PMID 12763985), a threshold inside most labs' normal range. A 2026 randomised placebo-controlled trial in non-anaemic iron-deficient women found the same direction (PMID 41736325).

Correct what is low, retest, and do not supplement speculatively expecting an energy effect.

Tier 2: studied in this population, with limits

Coenzyme Q10, with or without NADH

What the research shows. A prospective, randomized, double-blind, placebo-controlled trial of CoQ10 plus NADH in myalgic encephalomyelitis and chronic fatigue syndrome reported benefit for perceived cognitive fatigue and health-related quality of life, describing the combination as generally safe while calling for further work to confirm the benefit and clarify mechanism (PMID 34444817). A randomized, double-blinded, placebo-controlled study of ubiquinol-10 at 150 mg daily for 12 weeks in chronic fatigue syndrome found improvement in several symptoms including autonomic and cognitive measures (PMID 27125909).

How to read it. Two small placebo-controlled trials in an overlapping population, both positive, neither large. The population is ME/CFS rather than post-viral specifically, which is a real caveat and also the closest match available.

Typical studied range. 150 mg daily of ubiquinol-10 for 12 weeks (PMID 27125909).

Cautions. CoQ10 can reduce the effect of warfarin. It may modestly lower blood pressure, which matters if orthostatic symptoms are part of your picture. Not established in pregnancy.

Antihistamines, which belong here even though they are not supplements

Why they are in this guide. Because a lot of readers of this kit are quietly reacting to everything and have not connected it.

What was reported. In a study of long COVID patients with cardiovascular manifestations attributed to mast cell activation, antihistamines improved cardiovascular manifestations and other symptoms, with no significant change in the control group, which the authors take as supporting a role for mast cell activation in long COVID (PMID 37529714).

The limits, stated clearly. Small, not randomized, no placebo arm. That is enough to make an over-the-counter antihistamine trial a reasonable conversation with your clinician if reactivity is prominent. It is not enough to diagnose mast cell activation syndrome, and the consensus criteria for that diagnosis are strict, with a critical review finding that the alternative criteria in circulation often lack specificity and validation and are producing misdiagnosis (PMID 34166845).

Probiotics and the gut, honestly graded

A review of pharmacological and non-pharmacological management of long COVID discusses probiotics among approaches aimed at gastrointestinal health and overall wellbeing, and concludes that **large-scale randomized controlled trials are warranted to validate probiotic efficacy in long COVID** (PMID 41862909). A case-based systematic review of brain-gut axis approaches in long COVID and ME/CFS notes shared features including post-exertional malaise and gastrointestinal dysfunction, and calls for larger mechanistic studies and rigorously designed trials (PMID 41668172).

Translation. Plausible, actively researched, not yet demonstrated. Reasonable to try if gut symptoms are prominent. Not a core recommendation.

Tier 3: ask your prescriber first

- **Salt loading**, which is genuinely useful in orthostatic intolerance and genuinely inappropriate in hypertension, heart failure, and kidney disease.
- **Anything stimulant-like.** In a population where overexertion causes crashes, a product that makes you feel able to exceed your envelope is not a benefit.
- **High-dose antiviral botanical protocols marketed for EBV.** The lab guide explains why a positive EBV IgG is not the diagnosis these protocols assume.
- **Anything at all in pregnancy or nursing.**

What I deliberately left out, and why

- **Antiviral supplement protocols aimed at “chronic EBV.”** They rest on a misreading of the serology. Roughly nine in ten adults carry the antibody pattern being treated, and a positive VCA IgG with EBNA-1 IgG indicates past infection (PMID 28661055). Genuine chronic active EBV disease is rare, severe, and diagnosed in specialist settings.
- **Repeat EBV titres to track progress.** They will not track anything useful and they cost money every time.
- **High-dose immune “boosters.”** In a condition characterized partly by immune dysregulation, more stimulation is not obviously the goal, and there is no outcome evidence.
- **Graded exercise programs sold as recovery protocols.** Where post-exertional malaise is present this is the intervention most likely to set you back.
- **Anything promising a timeline.** Nobody can honestly tell you how long this takes.

Telling you what to skip is part of what you paid for, and here it also protects your energy budget.

Running a fair trial

Weeks 1 to 4, baseline. Change nothing. Score energy out of ten, note crash days, and record what you did the day before each crash.

Weeks 5 to 16, one intervention, at half the usual starting dose for the first week.

Week 17, judge it. In this condition, **fewer crashes is a better outcome signal than more energy**, and it usually moves first.

Bring the tracker. Months of daily scores is what makes a clinician useful to you.

Realistic expectations

What is realistic. Correcting a real deficiency helps over two to three months. Pacing reduces crash frequency. Orthostatic measures make being upright more tolerable. Those are real gains.

What is not realistic. A supplement resolving post-viral illness. There is no product that does that, and the honest state of the field is that treatment is not settled, which is why the recent literature calls for rehabilitation pathways and multidisciplinary care (PMID 42428921).

Timeline. Judge over months. Expect a non-linear course with good and bad stretches.

Cost sense. If something has done nothing after twelve weeks, stop paying for it.

When to seek professional care

Emergency. Chest pain, new or worsening breathlessness, coughing blood, swollen painful calf, fainting, one-sided weakness, trouble speaking, sudden severe headache.

Soon. Unintentional weight loss, persistent fever or night sweats, steadily worsening rather than fluctuating symptoms, fainting on standing, or a markedly changed resting heart rate.

Now: thoughts of harming yourself. Call or text 988.

Where to find quality versions of these

Nothing here names a brand, on purpose.

Look for third-party testing. NSF, USP Verified, or equivalent.

Match the form to the trial. Ubiquinol-10 at 150 mg is what was studied.

Read the whole panel. Fatigue and “immune” blends frequently hide stimulants, which is the last thing you want here.

If you would rather not evaluate all of that yourself, I keep a curated dispensary of professional-grade products at Root Health, screened for the standards above.

Root Health dispensary: us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the highest-value items in this guide, pacing and the standing test, cost nothing at all.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your history, and two questions that change everything: whether post-exertional malaise is present, and what happens to your heart rate when you stand.

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References

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Dr. Adam Seay, DC · Seay Wellness / Root Health, Garner, NC
rootcauseunlocked.com · (919) 662-0520

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